

Amazon RDS PostgreSQL Multi-AZ Instance Setup with Automatic Failover

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon RDS** PostgreSQL database was created using the **Dev/Test** template to enable a **Multi-AZ** deployment instead of the earlier single-AZ free-tier setup. The lab explained that the primary database instance runs in one Availability Zone while a non-readable standby instance is created in another Availability Zone for high availability. After selecting the engine version, instance class, storage, VPC, and security group, the database was created and then verified from the console. Once the status became available, the configuration showed **Multi-AZ = Yes** and displayed both the primary and secondary zones. The demonstration confirmed that if the primary Availability Zone fails, AWS automatically fails over to the standby instance to avoid downtime.

Understanding Amazon RDS Multi-AZ in PostgreSQL

Amazon RDS Multi-AZ keeps a database highly available by maintaining a primary instance and a standby instance in different Availability Zones. It exists so that if the primary AZ has a hardware, network, storage, or maintenance-related failure, the standby instance can take over automatically. The standby is not a read replica, so it cannot be queried directly; it is reserved purely for failover. This design is important for production databases because applications depend on the database staying online even when one AZ becomes unavailable.

Example:

A PostgreSQL database for an application is running in Availability Zone A, and its standby copy is continuously synchronized in Availability Zone B. If AZ A goes down because of a network or hardware issue, AWS automatically points traffic to the standby database in AZ B. The application keeps working without manual intervention or downtime.

Lab Steps

- Sign in to the AWS Management Console.
- Open **Amazon RDS**.
- Below Create with full configuration and Click **Create**.

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Aurora and RDS > Dashboard

Welcome to Aurora and RDS

Amazon Relational Database Service (Amazon RDS) and Amazon Aurora are fully managed database services that simplify setup, operation, and scaling of relational databases in the cloud.



Create with express configuration in seconds
Create and query an Aurora PostgreSQL serverless database with pre-configured settings to get started quickly.

Create



Create with full configuration
Create a database with custom configurations and enable features you want across a variety of engines including Aurora (compatible with MySQL & PostgreSQL), MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server, and IBM Db2. Or use an existing database using restore from S3.

Create Restore from S3

Service overview

Viewing data from Asia Pacific (Mumbai) (ap-south-1) Region.

DB clusters	DB instances	Snapshots	Recent events
0	0	5	0

[View service quota](#)

Service health

Current status | Details

[View service health dashboard](#)

Explore Aurora & RDS

In this activity, you will learn how to create a database. To begin, choose **Start tutorial**.

Estimated duration: 2-5 minutes

[Start tutorial](#)

RDS costs

- Select any engine but for this lab I have chosen **PostgreSQL**.

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Aurora and RDS > Databases > Create database

Availability zone group

Availability zone group
Select the availability zone where you want to create your database.

US East (N. Virginia)

Engine options

Engine type

☐ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☐ MySQL

☒ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Choose a database creation method

☒ Full configuration
Full configuration provides the most configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

- Choose **Full Configuration**.
- Pick the **Dev/Test** template.
- Choose **Multi-EZ DB Instance Deployment (2 Instance)**.

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Aurora and RDS > Databases > Create database

Templates

Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☒ Dev/Test
Intended for development use outside of a production environment.

☐ Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. [AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the Amazon RDS service level agreement \(SLA\)](#)

☐ Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency

☒ Multi-AZ DB instance deployment (2 instances)
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

- 99.95% uptime
- Redundancy across Availability Zones

☐ Single-AZ DB instance deployment (1 instance)
Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy

- Choose correct PostgreSQL Version.
- Enter the database identifier and credentials.

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Aurora and RDS > Databases > Create database

Engine version

View the engine versions that support the following database features.

▼ Hide filters

☐ Show only versions that support the Multi-AZ DB cluster [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

Engine version
PostgreSQL 18.2-R1

☐ Enable RDS Extended Support [Info](#)
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for PostgreSQL documentation](#).

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
database-2

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
postgres

The master username must be 1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
You must create a password or have RDS create a password that you manage.

- Select a suitable instance class and storage size but for lab purpose we are selecting lower ones as shown.

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Aurora and RDS > Databases > Create database

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs • 4 GiB RAM • EBS Bandwidth: Up to 2,085 Mbps • Network: Up to 5 Gbps

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)

Performance scales independently from storage

Allocated storage [Info](#)

20 GiB

Minimum: 20 GiB. Maximum: 6,144 GiB

Provisioned IOPS [Info](#)

3000 IOPS

- Choose the default VPC and attach the configured security group.
- Keep **Public access** set to **No** for the Multi-AZ setup.

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Aurora and RDS > Databases > Create database

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-099792461c7d0a616)

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-099792461c7d0a616

Public access [Info](#)

☐ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No

RDS can't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

Existing VPC security groups

Choose one or more options

EC2-SG default

- Leave the rest settings to default and create the database and wait for it to become available.
- Verify that the status changes from **creating/modifying** to **available**.
- Open the database configuration details.

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Aurora and RDS > Databases > database-2

database-2 [Modify] [Actions]

Summary

DB identifier database-2	Status Backing-up	Role Instance	Engine PostgreSQL	Recommendations
CPU 7.67%	Class db.t3.micro	Current activity 0.00 sessions	Region & AZ us-east-1b	

Connectivity & security | Monitoring | Logs & events | **Configuration** | Zero-ETL integrations | Maintenance & backups | Data migrations | Tags

Instance

Configuration DB instance ID database-2 Engine version 18.2 RDS Extended Support Disabled DB name - License model Postgresql License	Instance class Instance class db.t3.micro vCPU 2 RAM 1 GB Availability Master username postgres	Primary storage Encryption Enabled Encryption key AWS managed KMS key aws/rds Storage type General Purpose SSD (gp3) Storage 20 GiB Provisioned IOPS	Monitoring Monitoring type Database Insights - Standard Performance Insights Enabled Retention period 7 days AWS KMS key aws/rds Enhanced Monitoring Enabled
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- Confirm that **Multi-AZ** is enabled.

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Aurora and RDS > Databases > database-2

Connectivity & security | Monitoring | Logs & events | **Configuration** | Zero-ETL integrations | Maintenance & backups | Data migrations | Tags

Instance

Configuration DB instance ID database-2 Engine version 18.2 RDS Extended Support Disabled DB name - License model Postgresql License Option groups default:postgres-18 In sync Amazon Resource Name (ARN) arn:aws:rds:us-east-1:463646775279:db:database-2 Resource ID db-QCSUJUVNBVYVH47XQSTUISOVBBY Created time	Instance class Instance class db.t3.micro vCPU 2 RAM 1 GB Availability Master username postgres Master password ***** IAM DB authentication Not enabled Multi-AZ Yes Secondary Zone us-east-1d	Primary storage Encryption Enabled Encryption key AWS managed KMS key aws/rds Storage type General Purpose SSD (gp3) Storage 20 GiB Provisioned IOPS 3000 IOPS Storage throughput 125 MiBps Storage autoscaling Enabled Maximum storage threshold 1000 GiB Storage file system configuration	Monitoring Monitoring type Database Insights - Standard Performance Insights Enabled Retention period 7 days AWS KMS key aws/rds Enhanced Monitoring Enabled Granularity 60 seconds Monitoring role arnaws:iam:463646775279:role/rds-monitoring-role
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- Verify that the primary zone and secondary standby zone are different.

The screenshot displays the AWS Management Console for an Amazon RDS database instance named 'database-2'. The interface is divided into a left-hand navigation pane and a main content area. The navigation pane includes links for 'Aurora and RDS', 'Databases', and various management tools like 'Query editor', 'Performance insights', and 'Snapshots'. The main content area shows the 'Summary' and 'Configuration' tabs for the instance. The 'Summary' tab highlights the instance's status as 'Available', its role as 'Instance', and its engine as 'PostgreSQL'. The 'Configuration' tab provides detailed settings for the instance, including its class, storage, and monitoring options. A red box highlights the 'Region & AZ' field, which is set to 'us-east-1b'.

Summary			
DB identifier database-2	Status Available	Role Instance	Engine PostgreSQL
CPU -	Class db.t3.micro	Current activity	Region & AZ us-east-1b

Instance			
Configuration DB instance ID database-2 Engine version 18.2 RDS Extended Support Disabled DB name - License model Postgresql License	Instance class Instance class db.t3.micro vCPU 2 RAM 1 GB Availability Master username postgres	Primary storage Encryption Enabled Encryption key AWS managed KMS key aws/rds Storage type General Purpose SSD (gp3) Storage 20 GiB Provisioned IOPS	Monitoring Monitoring type Database Insights - Standard Performance Insights Enabled Retention period 7 days AWS KMS key aws/rds Enhanced Monitoring Enabled

- Confirm that the standby database is non-readable and exists only for failover.

Verification

- The PostgreSQL RDS instance was created successfully.
- The database was deployed in **Multi-AZ** mode.
- The primary and standby instances were placed in different Availability Zones.
- The console confirmed **Multi-AZ = Yes**.
- The setup was verified as an automatic failover architecture for high availability.